Overview of Research Agenda

As a researcher, my interests revolve around utilizing technology to improve undergraduate education. Before starting at ERAU, I began using programming languages to build automated student-to-expert interactions that are critical to improving student learning. I have since expanded my computer skills to include machine learning through a 5-course Coursera specialization. I am now uniquely positioned as a multi-disciplinary expert in mathematics, mathematics education, and data science.

At the macro learning-level, I study how different types of educational activities (e.g., discussions, quizzes, exploratory activities, digital manipulatives) may impact student understanding. At the micro learning-level, I create models that predict how students may respond to individual questions that are then used to provide automated feedback. My wide range of educational research, statistics, and programming skills have also led to productive collaborations where I act as an expert in data collection and qualitative/quantitative data analysis to improve undergraduate learning. My active research projects reflect these varied data science skills that all revolve around improving online education for college students.

Given the numerous successful funding proposals I have been a part of, associated with different research projects and collaborators, I have organized my research experiences by topic and summarized how they fit within my research agenda.

Generative AI in Education

Generative AI became a massive disruption to education, especially online education, in early 2023. I have since become increasingly involved in research on Generative AI's impact in education.

Generative AI Discussion Feedback: In Fall 2023, Alex Rister organized a cohort of faculty from various COAS-WW departments to investigate the extent to which Generative AI could provide feedback to student discussion responses. The team was awarded an internal grant and collected data from several courses. As part of this project, I contributed to 1 national conference presentation, 1 local conference presentation, and 2 accepted journal articles.

Generative AI in Writing Tasks: In Fall 2024, I began working with Emily Dux Speltz on Generative AI use in both academic and professional writing. We received an ERAU FIRST grant in Spring 2025 through which we have interviewed professionals and students after first recording their use of Generative AI to complete a professional or educational writing task. As part of this project, I have contributed to coordinating the 3 student research assistants on the grant and 2 international conference presentations. My prior research with Meghan Velez (University of Central Florida) has also led to a cross-institution NSF grant submission between UCF, ERAU, and USF to conduct a longitudinal study on student use of Generative AI in writing assignments, though it was not funded at this time.

Generative AI in Mathematics Education: As part of my responsibilities as chair of the Mathematical Association of America (MAA) Subcommittee on Technology in Mathematics Education, I have advised various MAA committees on the potential impacts of Generative AI in mathematics education. This led to leading a submission of an external Spencer Grant with MAA that would create an invite-only conference to discuss and spearhead research on Generative AI in Mathematics Education. After this Spencer Grant was declined, it was subsequently submitted to NSF and awarded in May 2026. The conference took place in June 2026 and produced a statement on Generative AI in Mathematics Education that has been approved by the MAA Board of Directors: https://maa.org/resource/statement-on-ai-in-higher-education/.

Educational Activities in Online STEM Education

While Generative AI has tasks it can do well (summarizing materials, translating languages, and small-scale coding tasks), it is not yet capable of providing nuanced feedback to student work without error. Thus I have been exploring alternative machine learning methods to detect patterns in student thinking and provide automated feedback. This has led to two threads of related activity: creating autonomous educational assessments for student learning and autonomously evaluating assessments for instructor review.

Autonomous Feedback to Students: Before joining ERAU-WW, I created a mastery-based College Algebra course at the University of Florida. Unlike a traditional course where content is presented on a schedule regardless of a student's performance, mastery-based courses have students focus on a small number of learning objectives until they display mastery, then allow the student to continue on to new learning objectives. In Fall 2023 I received a small external grant to implement a new educational technology (DOENET) to create a dynamic activity for inclusion in ERAU-WW's College Algebra course. Like my prior work at UF, I identified patterns of student thinking for inclusion in a College Algebra word problem. However, I was able to leverage the dynamic capabilities of DOENET to display and hide content on the webpage based on how a student responded. This dynamic activity was thus able to act as a "Choose Your Own Adventure" where students would follow paths of thinking, correct or incorrect, until it prompted them to return to some critical juncture with additional feedback on how they could change their thinking. In collaboration with colleagues at Georgia State University (GSU), we submitted a Collaborative Proposal to NSF in Fall 2024 and Fall 2025 to develop these types of dynamic activities as well as the autonomous feedback homework type assignments for GSU's Calculus 1 courses. The proposal was reviewed as Competitive each time but was not funded. This work has produced 1 journal article under review, 1 published journal article, 2 conference proceedings papers, 6 national conference presentations, and 2 local conference presentations.

Conceptual Understanding in Asynchronous Discussions: In Spring 2023 I led another internal SEED grant with Karen Keene and Zachary Reed to investigate how students' conceptual understanding progressed in asynchronous discussions in Calculus. We eventually published a methodology for identifying conceptual understanding and its progression in asynchronous discussions. This collaborative project was cut short after the untimely passing of Karen Keene in February 2023. These two SEED grants produced 1 book chapter, 2 conference proceedings papers, 2 national conference presentations, 1 regional conference presentation, and 1 local conference presentation.

Autonomous Feedback to Instructors: My efforts with autonomous feedback to instructors led to Emily Faulconer and Beverly Wood asking me to join their NSF grant when I started at ERAU in Fall 2021. In this project I worked with undergraduate student research assistants to qualitatively code student cognitive and social moves in online discussions. We then began extracting features from student discussion posts that would allow for autonomous qualitative coding and thus autonomous reports to instructors to provide top-level summaries of student participation in discussions. As Generative AI was quite proficient at summarizing by the end of 2024, the goal of automating qualtiative coding became moot. The project did provide insights on students' cognitive and social moves in a course though and was productive, producing 2 journal articles, 1 conference proceeding, 2 national conference presentations, 1 invited talk, 3 white papers, and 2 editorials.

Data Analysis for Engineering Education Projects

Since my expertise in data science can be applied to various fields outside of my immediate research interests, I continue to provide data science support for research projects led by ERAU-WW Engineering faculty. This allows me to support ERAU's signature research fields of aviation and aerospace education with my expertise in educational data science.

Research Scholars Program: In Spring 2023, I was approached by engineering faculty Robert Deters and Brent Terwilliger to act as the data scientist on their continuing submission of the Research Scholars Program. This project focuses on developing resources for supporting undergraduate research in online programs. This second stage expanded the impact to all ERAU-WW students, helped develop a cross-campus Undergraduate Research minor, expanded on resources to support undergraduate research, and studied the effects of these efforts. This project produced 1 journal article under review, 3 conference proceedings, 5 national conference presentations, 1 white paper, and 24 workshops.

Autonomous Curriculum Suggestions: In Fall 2025, I was approached by engineering faculty Ghazal Barari and Robert Deters to utilize artificial intelligence and machine learning to support curriculum updates based on curriculum signals from the workforce, such as skills detailed in job calls. This NSF proposal was submitted in November 2025 and deemed Competitive but not funded. It has since been revised and resubmitted to another NSF program in October 2026.